

Ubuntu Static IP Configuration Guide

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Linux System Administration — Network Configuration

1. Overview

This document provides step-by-step instructions for assigning a static IP address on Ubuntu systems using Netplan, the default network configuration tool since Ubuntu 18.04.

A static IP ensures a machine always uses the same address on the network. This is essential for servers, printers, network devices, and any system that other hosts need to reach reliably.

2. Prerequisites

Before proceeding, make sure you have the following:

- Ubuntu 18.04 or later (Netplan is the default network manager)
- Root or sudo privileges on the target machine
- Your desired static IP address and subnet mask (e.g. 192.168.1.100/24)
- Your network gateway IP address (e.g. 192.168.1.1)
- Preferred DNS server addresses (e.g. 8.8.8.8, 1.1.1.1)
- A terminal and a text editor such as nano or vim

3. Step-by-Step Configuration

Step 1 — Find Your Network Interface Name

Run one of the following commands to list all network interfaces on the system:

```
ip link show  
  
# or alternatively:  
ip a
```

Look for your Ethernet interface in the output. Common interface names include:

- eth0 — traditional naming on older systems

- ens33 / ens3 — common on VMware and cloud VMs
- enp3s0 — PCI-based physical interfaces

Step 2 — Locate the Netplan Configuration File

Netplan configuration files are stored in `/etc/netplan/`. Check what files exist:

```
ls /etc/netplan/
```

The filename typically looks like `00-installer-config.yaml` or `01-netcfg.yaml`. Back up the file before editing:

```
sudo cp /etc/netplan/00-installer-config.yaml ~/netplan-backup.yaml
```

Step 3 — Edit the Configuration File

Open the Netplan configuration file in a text editor:

```
sudo nano /etc/netplan/00-installer-config.yaml
```

Replace or update the contents with the following structure. Substitute all values to match your actual network environment:

```
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: false           # Replace with your interface name
                             # Disables automatic IP assignment
      addresses:
        - 192.168.1.100/24   # Your static IP and subnet mask
      routes:
        - to: default
          via: 192.168.1.1    # Your gateway (router) IP
      nameservers:
        addresses:
          - 8.8.8.8          # Primary DNS
          - 1.1.1.1          # Secondary DNS
```

IMPORTANT: YAML files are indentation-sensitive. Always use spaces, never Tab characters. Incorrect indentation will cause Netplan to fail or silently ignore the configuration.

Step 4 — Test and Apply the Configuration

Before fully committing the change, test it first. Netplan will automatically roll back if you do not confirm within 120 seconds:

```
sudo netplan try
```

If the network connection is still working and you are satisfied, press Enter to confirm. Then apply permanently:

```
sudo netplan apply
```

Step 5 — Verify the Configuration

Confirm the static IP address has been applied and the network is reachable:

```
# Show the IP address of your interface
ip a show ens33

# Verify the default gateway route
ip route

# Test connectivity to the internet
ping -c 4 8.8.8.8

# Test DNS name resolution
ping -c 4 google.com
```

4. Subnet Mask Reference

Use this table to identify the correct CIDR notation for your network subnet:

CIDR	Subnet Mask	Usable Hosts	Example Range
/24	255.255.255.0	254	192.168.1.1 – .254
/23	255.255.254.0	510	192.168.0.1 – 1.254
/22	255.255.252.0	1,022	192.168.0.1 – 3.254
/16	255.255.0.0	65,534	10.0.0.1 – 255.254
/8	255.0.0.0	16,777,214	10.0.0.1 – 255.255.254

5. Tips and Best Practices

- Choose a static IP that is outside the DHCP pool of your router to prevent address conflicts.
- Always use `sudo netplan try` before `sudo netplan apply` so changes roll back automatically if something goes wrong.
- Back up your Netplan config file before every edit.
- On Ubuntu Desktop, static IP can also be set via Settings > Network > gear icon > IPv4 tab.
- For Wi-Fi interfaces, replace the `ethernets: block` with a `wifis: block` in your Netplan config.
- If managing multiple interfaces, add each one as a separate entry under `ethernets:`.

6. Troubleshooting

Changes not taking effect

If the static IP does not appear after running `netplan apply`, check for YAML syntax errors by generating the config:

```
sudo netplan generate
```

Read the output carefully. Any indentation or syntax error will be shown here.

Lost SSH or remote access after applying

If you lose remote access, you likely entered an incorrect IP or gateway. Use a local console (KVM, iDRAC, or VM console) to temporarily restore DHCP and reconnect:

```
sudo dhclient ens33      # Request IP via DHCP
ip a show ens33          # Confirm the DHCP address
```

Interface name not found in config

Interface names can differ between machines or change after hardware updates. Always confirm the exact interface name with `ip link show` before editing the Netplan file.

DNS not resolving after static IP

If `ping 8.8.8.8` works but `ping google.com` fails, DNS is not configured properly. Verify the nameservers block in your Netplan config and re-apply. Also check:

```
cat /etc/resolv.conf
resolvectl status
```

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